

Carl McMillan

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Objective

My background is mostly as a software developer and devops engineer, I am currently studying AI as I believe it can provide very useful tools to help us with everyday tasks, I am driven by solving problems in simple and efficient ways.

Programming Languages

Java, Python, Groovy, SQL, Javascript, Chef/Ruby, HCL

- Used Java for creating bigger applications and microservices, mostly for integration, main reason I like it is extensive support and libraries
- Use Python for machine learning and where scripts are needed but also like it as an application language as it is simple and also has good libraries
- Have used SQL and Javascript a lot back when i was making web apps, but not as much now as have been doing mainly integration
- Used Chef/Ruby for building VM's and installing applications where Docker wasn't an option, but use Docker/Kubernetes where I can
- Use HCL (Hashicorp configuration language) for building environments in AWS
- Often used languages like Groovy for working with different tools like Jenkins

Tools/Platforms

AWS, Terraform, Jenkins, Gitlab, Linux, Git, Gradle, Maven, Docker

Work Experience

Research Assistant at Victoria University of Wellington

April 2023 - June 2023

- Annotated a mussel float detection and buoyancy dataset used by multiple researchers for the Centre for Data Science and Artificial Intelligence aquaculture research group at VUW
- Produced a video demonstration by training an object detection model on this dataset to showcase the capabilities of the Centre for Data Science and Artificial Intelligence

Devops engineer at Fronde

Sept 2020 - Sept 2022

- Building resources in AWS for integrated microservice environments using terraform
- CI/CD using Gitlab CI to build pipelines for monorepo microservices project
- Monitoring and alerting using AWS Cloudwatch, Prometheus, Grafana, and Slack integrations

Technical Consultant (development/devops) at Westpac

June 2017 - Sept 2020

- Building java microservices in Kubernetes using Kafka for distributed messaging, used for integration between proprietary bank card processing software running on older platforms.
- Development using encryption for secure storage of data in database's and Kafka topic's for PCI DSS compliance for financial transactions.
- Automation of building servers and installation of proprietary bank card processing software using Chef and Jenkins.
- Monitoring and alerting of the mentioned systems using Elasticstack, Grafana, PagerDuty.
- On call production support for the above systems.

Software Developer at Datacom

June 2011 - June 2017

- Developing web data visualisation software for tracking availability of Chorus resources.
- Part of development team for integration components of ordering system for Chorus fibre.
- Integration development for AMS for remote operations on smart meters.
- On call production support for existing apps in our team.
- Improvements to our teams development processes, e.g. automation of processes, chat ops.

Software Developer at Wialus

Nov 2009 - Feb 2010

- Summer of Code internship.
- Created C++ programs for ETL processes for mobile phone signal data.
- Created C++ program for calculating elevations along any great circle path.

Biosecurity Officer at GWRC**Jul 2006 - Nov 2007**

- Control pest animals in the Wellington region, and help people with pest animal problems.
- Set up poison and trapping programs for volunteers to take over.
- Mapping with GPS devices and ArcGIS software.
- Looking after database and generated reports for a large conservation project in Wainui water catchment.
- Conducted trials for possum control methods.

Education

Victoria University**In Progress: Submitted**

- PhD in Artificial Intelligence

Victoria University**Completed Oct 2023**

- Postgraduate Diploma in Science in Artificial Intelligence

Victoria University**Completed Dec 2011**

- Conjoint BSc/BCA in Computer Science/Mathematics/Money & Finance

Victoria University**Completed May 2004**

- BSc in Ecology & Biodiversity/Marine Biology